

Task: 01

Create a program that creates three threads. Each thread should perform a different task:

- Thread 1: Print numbers from 1 to 10.
- Thread 2: Print the first 10 Fibonacci numbers.
- Thread 3: Print the first 10 prime numbers.

Task: 02

Write a program that uses shared memory to aggregate data from multiple processes. The goal is to calculate the sum of an array of integers, where each process computes a partial sum of a subset of the array and stores it in a shared memory segment. The main process will then read the partial sums from the shared memory and compute the final result.

Details:

1. Define a shared memory segment large enough to store:
 - An array of integers (size N).
 - An array of partial sums (size M, where M is the number of processes).
 - A final sum variable.
2. The main process should:
 - Create the shared memory segment.
 - Initialize the array of integers with random values (using rand()).
 - Fork M child processes.
3. Each child process should:
 - Compute the sum of a subset of the array (divide the array into M equal parts).
 - Write its partial sum into the shared memory segment at the appropriate index.
 - Exit after writing the partial sum.
4. The main process should:
 - Wait for all child processes to complete.
 - Read the partial sums from the shared memory and compute the final sum.
 - Print the final sum.
 - Detach and delete the shared memory segment.